

Expanding Brackets

Learning Objectives

- Simplify algebraic expressions
- Expand brackets
- Solve equations in which the unknown appears on both sides

Basic Principles

Algebra uses letters, often x , to stand for numbers. Algebraic expressions can be treated in the same way as number expressions.

- $x + 3$ means “add three to the unknown number.”
- $3x$ means “3 times the unknown number.”
- x^2 means “square the unknown number.”

Example: Combining Like Terms

$$a + 3ab - 4ba = a - ab$$

i Note

Note: $ab = ba$, so $3ab$ and $-4ba$ are like terms.

$$3p^3 + 2p^2 - 2p^3 + 5p^2 = (3p^3 - 2p^3) + (2p^2 + 5p^2) = p^3 + 7p^2$$

Example: Multiplying Algebraic Terms

$$4r \times 5t = 20rt$$

$$(3b)^2 \times 3b = 9b^2 \times 3b = 27b^3$$



Tip

- The multiplication sign is often not included between letters, e.g. $3ab$ means $3 \times a \times b$
- When multiplying, add like powers: $3a^2b \times 2a^5b^4 = 6a^8b^5$. (Think of a as a^1).

Example: Expanding Brackets

$$2(3 + x) = 2 \times 3 + 2 \times x = 6 + 2x$$

Tip

- Multiply each term inside by the outside term.
- The multiplication sign is usually left out: $3(x + y)$ means $3 \times (x + y) = 3 \times x + 3 \times y = 3x + 3y$
- When multiplying, the number 1 is usually left out: $-(2x+3)$ means $-1 \times (2x+3) = (-1) \times (2x) + (-1) \times 3 = -2x - 3$

 Warning

Be very careful with negative signs outside a bracket.

$$-2 \times (a - 3) = (-2) \times a + (-2) \times (-3) = -2a + 6$$

Example: Expanding with Indices

Expand $5x^2(12x^3 + 4x^5)$

$$5x^2 \times 12x^3 + 5x^2 \times 4x^5 = 60x^5 + 20x^7$$

Test Your Understanding

Expand

1) $4y(2y - 8)$

2) $3x^2(x^3 + y^4)$

3) $c^{\frac{1}{2}}(c^{\frac{3}{2}} - c^{\frac{1}{2}} + c^{-\frac{1}{2}})$

4) $5x^2(6x - x^2 + 7y - x^{-2})$

Answers

$$1) 4y(2y - 8) = 8y^2 - 32y$$

$$2) 3x^2(x^3 + y^4) = 3x^5 + 3x^2y^4$$

$$3) c^{\frac{1}{2}}(c^{\frac{3}{2}} - c^{\frac{1}{2}} + c^{-\frac{1}{2}}) = c^2 - c + 1$$

$$4) 5x^2(6x - x^2 + 7y - x^{-2}) = 30x^3 - 5x^4 + 35x^2y - 5$$

Exercise 1

Please complete the worksheet

Example: Expanding Double Brackets

To expand $(x + 3)(x + 2)$, you can use the grid method.

	x	$+2$
x	x^2	$2x$
$+3$	$3x$	6

Summing the terms on the inside and simplifying gives:

$$x^2 + 3x + 2x + 6 = x^2 + 5x + 6$$

Example: Squared Bracket

A squared bracket is simply the same bracket twice. For example to expand

$$(2x + 3)^2$$

the grid method is:

	$2x$	$+3$
$2x$	$4x^2$	$6x$
$+3$	$6x$	9

Adding the terms on the inside and simplifying gives

$$4x^2 + 12x + 9$$

Test Your Understanding

Expand and simplify:

1) $(2y + 3)^2$

2) $(4 + x)(3x - 1)$

3) $(4a + 3)(4a - 3)$

4) $(5 - 2a)(b - 6)$

Answers

$$1) (2y + 3)^2 = 4y^2 + 12y + 9$$

$$2) (4 + x)(3x - 1) = 3x^2 + 11x - 4$$

$$3) (4a + 3)(4a - 3) = 16a^2 - 9$$

$$4) (5 - 2a)(b - 6) = 5b + 12a - 2ab - 30$$

Exercise 2

Please complete the worksheet

Example: 2×3 Grids

Expand and simplify $(x^2 + 6x - 3)(x - 4)$

	x	-4
x^2	x^3	$-4x^2$
$6x$	$6x^2$	$-24x$
-3	$-3x$	12

$$\begin{aligned} & x^3 + 6x^2 - 3x - 4x^2 - 24x + 12 \\ &= x^3 + 2x^2 - 27x + 12 \end{aligned}$$

Test Your Understanding

Expand and simplify:

1) $(a - 3b + 2c)(a + b)$

2) $(cd + 4c - 5d)(3e + 2f)$

3) $(x^2 - x + 7)(2x + 3)$

4) $(1 + 3x - 2x^2)(7x - 2)$

Answers

$$1) (a - 3b + 2c)(a + b) = a^2 - 2ab + 2ac + 2bc - 3b^2$$

$$2) (cd+4c-5d)(3e+2f) = 3cde+2cdf+12ce+8cf-15de-10df$$

$$3) (x^2 - x + 7)(2x + 3) = 2x^3 + x^2 + 11x + 21$$

$$4) (1 + 3x - 2x^2)(7x - 2) = -14x^3 + 25x^2 + x - 2$$

Exercise 3

Please complete the worksheet.

Example: Triple Brackets

A three-bracket expansion such as $(x + 1)(x - 3)(x + 2)$ can be done in two grid-method steps. First, expand $(x - 3)(x + 2)$:

	x	$+2$
x	x^2	$2x$
-3	$-3x$	-6

This gives a result of

$$x^2 - x - 6$$

Then expand $(x + 1)(x^2 - x - 6)$ with a 2×3 grid:

	x^2	$-x$	-6
x	x^3	$-x^2$	$-6x$
$+1$	x^2	$-x$	-6

So altogether $(x + 1)(x - 3)(x + 2) = x^3 - 7x - 6$

 Tip

The order you multiply the brackets does not matter.

Test Your Understanding

Expand and simplify:

1) $(x + 2)^3$

2) $(2x + 1)(x - 1)^2$

3) $(3 + y)(3 - y)^2$

4) $(a - bx)^3$

Answers

$$1) (x + 2)^3 = x^3 + 6x^2 + 12x + 8$$

$$2) (2x + 1)(x - 1)^2 = 2x^3 - 3x^2 + 1$$

$$3) (3 + y)(3 - y)^2 = y^3 - 3y^2 - 9y + 27$$

$$4) (a - bx)^3 = a^3 - 3a^2bx + 3ab^2x^2 - b^3x^3$$

Exercise 4

Please complete the worksheet.